

# 16. DOM Tree

## Basic idea

HTML ka **har tag ek object** hota hai. Nested tags = "children". Tag ke andar ka text bhi object hota hai. JS se in sab ko access/modify kar sakte ho.

js

```
document.body.style.background = 'red';  
setTimeout(() => document.body.style.background = '', 3000);
```

Aur properties: `innerHTML`, `offsetWidth`, etc.

## DOM = Tree

html

```
<html>  
  <head><title>About elk</title></head>  
  <body>The truth about elk.</body>  
</html>
```

- Tags = **element nodes**, jo tree banate hain (`<html>` root, `<head>` aur `<body>` uske children).
- Tag ke andar ka text = **text node** (`#text`). Ye hamesha leaf hota hai (uske children nahi hote).

**Spaces aur newlines bhi text nodes bante hain!** Jaise `<head>` ke andar `<title>` se pehle ki khaali jagah.

Do exceptions:

1. `<head>` se pehle ke spaces ignore hote hain.
2. `</body>` ke baad kuch likhoge to wo `<body>` ke andar aakhir me chala jaata hai.

## Autocorrection

Galat HTML ko browser khud theek karke DOM banata hai:

- `<html>`, `<body>` na ho to bana deta hai.
- Unclosed tags (jaise `<li>`) band kar deta hai.
- Table me `<tbody>` hamesha add hota hai, chahe HTML me na likha ho.

## Baaki node types

- **Comment node** (`#comment`): comments bhi DOM me aate hain.

- `<!DOCTYPE>` bhi ek node hai.
- `document` object bhi formally ek node hai.

Total 12 node types hain, par 4 hi kaam aate hain:

1. `document` (entry point)
2. element nodes (tags)
3. text nodes
4. comment nodes

## DOM ko dekhna

- Browser DevTools > **Elements** tab (right-click > Inspect).
- Sub-tabs: **Styles** (applied CSS), **Computed** (final CSS per property), **Event Listeners**.
- Live DOM Viewer bhi ek tool hai.

## Console ke saath

- Elements me kuch select karo, phir console me `$0` us selected element ko deta hai (`$1` pichhla).

```
js
```

```
$0.style.background = 'red';
```

- Ulta: `inspect(node)` node ko Elements tab me dikhata hai.

## Yaad rakho

HTML ka har hissa (tags, text, comments) DOM me ek node hai. DevTools DOM samajhne ke liye best tool hai.